

Subject Description Form

Subject Code	ITC4212I
Subject Title	CAD for Intimate and Sports Fashion
Credit Value	3
Level	4
Pre-requisite(s) / <Co-requisite> / (Exclusion)	ITC3010I Bra Construction or ITC3218I Lingerie Development
Objectives	The subject provides in-depth theoretical and practical knowledge of CAD systems employed in the intimate apparel and activewear industry. It adopts CAD as a tool at all stages of the design and development process, including pattern construction, grading, marker planning and making.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - identify the purpose of CAD applications used in the intimate apparel and activewear industry in general; - analyse the functions of different tools in a CAD system; - apply the principles of CAD in pattern design, grading and marker making; - demonstrate an ability to incorporate CAD systems to facilitate intimate apparel and activewear pattern design, manipulation and development, sizing and grading, marker planning and making; - demonstrate the necessary communication skills to present pattern design ideas and concepts at a level acceptable to industry.
Subject Synopsis/ Indicative Syllabus	(I) Fundamentals of CAD System Production cycle and advantages in using CAD in intimate apparel and activewear production (II) Applications of Pattern CAD System in Intimate and Activewear Pattern design Digitizing Grading and sizing Marker making 3D garment design (III) Practical CAD Skills Development Basic pattern drafting for intimate and activewear Style manipulation Pattern grading and sizing Marker management 3D garment visualization

Teaching/Learning Methodology	This subject will be taught through a series of practical demonstrations in the computer laboratory. Students will be given different styles of intimate and activewear garment to create, manipulate and grade. It will be designed to develop students' critical thinking, analytical skills, teamwork, and oral and written communication skills.																																																											
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr> </thead> <tbody> <tr> <td>Continuous Assessment</td><td>100%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>1. Tests</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr> <tr> <td>2. Project</td><td>50%</td><td></td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>3. Class Participation</td><td>10%</td><td></td><td></td><td></td><td></td><td>✓</td></tr> <tr> <td>Examination</td><td>0%</td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>Total</td><td>100%</td><td></td><td></td><td></td><td></td><td></td></tr> </tbody> </table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Tests: Student's comprehension of CAD system used in the intimate and activewear industry will be assessed via a series of tests. Project: It will be a learner-centred and task-oriented project for assessing intermediate and final learning outcomes. Student's pattern development skills will be assessed via the project. The progress of the students will be monitored throughout the different stages of the study. Class participation: Student's professionalism, including communication skills and punctuality, will be assessed via attendance and participation in class.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	Continuous Assessment	100%	✓	✓	✓	✓	✓	1. Tests	40%	✓	✓	✓			2. Project	50%		✓	✓	✓	✓	3. Class Participation	10%					✓	Examination	0%						Total	100%					
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Student Study Effort Expected	Class contact:																																																											
	▪ Studio					39 Hrs.																																																						
	Other student study effort:																																																											
	▪ Project					66 Hrs.																																																						
	Total student study effort					105 Hrs.																																																						

Reading List and References	<u>User Manual</u> • AGMS garments CAD system manual, ASAHI KASEI AGMS Corporation. • AccuMark PDS manual, Gerber Garment Technology. AccuMark Marker Making manual, Gerber Garment Technology. • PAD manual, Pattern Enterprise and Pattern Professional V6.0. PAD manual, <u>Marker Design V6.0 English Version</u>. PAD manual, <u>Digitizer Manager V6.0 English Version</u>. PAD manual, <u>Plot Manager V6.0 English Version</u>. • OptiTex manual OptiGrade manual OptiMark manual OptiNest manual OptiDigitize manual Runway Designer <u>Books</u> Cole, J. (2016), <i>Patternmaking with stretch knit fabrics</i>. Fairchild Books, an imprint of Bloomsbury Publishing Inc. Gray, A. (2006), <i>Modern differential geometry of curves and surfaces with Mathematica</i>. (3rd Ed.) CRC, Chapman & Hall. Haggar, A. (2004), <i>Pattern cutting for lingerie, beachwear and leisurewear</i>. (2nd Ed.) Iowa State Press, Blackwell Pub. Johnson, B. (2005), <i>The bra-makers manual</i>. (1st ed.) Turtle Press, Ontario. Shin, K. (2010), <i>Patternmaking for underwear design</i>. Createspace. Loehr, N. (2013), <i>Demystifying bra fitting and construction</i>. Orange Lingerie LLC. Matthews, J. L. (2012), <i>Bare essentials: bras: construction and pattern drafting for lingerie design</i>. Fairbanks Publishing, Los Angeles Fashion Resource. Matthews, J. L. (2011), <i>Bare essentials: underwear: construction and pattern drafting for lingerie design</i>. Fairbanks Publishing, Los Angeles Fashion Resource. 張鴻志, 趙銘平, 謝朝 (2002), <i>服裝紙樣計算機輔助設計 Garment pattern computer aided design</i>, 北京中國紡織出版社. 常建亮 (2008), <i>內衣工業：紙樣與工藝</i>. (第 1 版) 上海科學技術出版社, 上海.
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